

Analytic derivation for the large- n theorem

This derivation accompanies [paper.md](#). Its scope is precisely the coefficient signs and zero classification for $n \geq 31147$. The mathematical displays in §§1–10 retain the integrated analytic source; the surrounding status text has been updated for this edition. Finite coefficient runs below the threshold and unrelated historical diagnostics are omitted. Section numbers are retained for cross-reference to formulas. The [Lean supplement](#) supplies the formal dependency chain.

1. Claims, notation, and proof dependencies

Theorem A

For integers $n \geq 31147$ and $0 \leq m \leq 10n^2$,

$$c_n(m) \geq 0 \quad (5 \mid m), \quad c_n(m) \leq 0 \quad (5 \nmid m). \quad (1.1)$$

For the same range of n , the zero set is

$$\begin{aligned} Z_n^- &= \{5j + 3, 5j + 4 : 0 \leq j < n\} \cup \{7, 5n + 8, 5n + 9\}, \\ Z_n &= Z_n^- \cup (10n^2 - Z_n^-), \quad |Z_n| = 4n + 6. \end{aligned} \quad (1.2)$$

The following sections give the analytic argument. The supporting Lean development proves the large- n sign theorem and zero classification; independent human peer review is not claimed.

Stage	Content	Principal dependencies
§2–3	5-dissection of the infinite product, nonnegative kernel, and first-tail formula	classical identities, formal power series
§4–5	explicit relative error at the endpoint	eta transformation, tail expansion, all-angle gap
§6	endpoint region for $n \geq 31147$	§3, §5
§7–9	absolute error on the fixed interior interval	phase lower bound, localization of the finite product, Gaussian integral
§10	passage to all coefficients and the zero set	§6, §9, reciprocal symmetry

We do not assume the implicit eventual theorem from the original notes. Nor do we depend on the Round 3 theorem for all small primes; the interior estimate for $p = 5$ that is needed here is given in this manuscript.

Put $D = 10n^2$ and $\zeta = e^{2\pi i/5}$. Since there are $4n$ factors,

$$c_n(m) = c_n(D - m). \quad (1.3)$$

Inversion sends a residue class a to $-a \bmod 5$, while preserving the desired sign.

The basic functions are

$$\begin{aligned} h(z) &= \frac{\text{Li}_2(e^{-z}) - \pi^2/6}{z}, \\ R(z) &= h(5z) - h(z), \quad C = \frac{2\pi^2}{15}, \quad A = \frac{C}{5}. \end{aligned} \quad (1.4)$$

The branch of h is taken from the right half-plane. For real $\tau \geq 0$,

$$R(\tau) = \int_0^1 \log \left(\sum_{j=0}^4 e^{-j\tau x} \right) dx, \quad R(0) = \log 5, \quad R'(0) = -1, \quad R''(0) = \frac{2}{3}. \quad (1.5)$$

Using the weights $\Pr(J = j) \propto e^{-j\tau x}$ gives

$$R''(\tau) = \int_0^1 x^2 \text{Var}(J) dx > 0. \quad (1.6)$$

Thus $-R'$ decreases strictly from 1 to 0. If $0 < k < 5n^2$, then

$$-R'(\tau) = \frac{k}{5n^2} \quad (1.7)$$

has a unique positive solution. The case $k = 5n^2$ is treated at $\tau = 0$.

2. The 5-dissection of the infinite product and the zero residue classes

Define $(a; Q)_j = \prod_{r=0}^{j-1} (1 - aQ^r)$. Then

$$G(q) = \frac{(q; q)_\infty}{(q^5; q^5)_\infty}, \quad T_n(q) = \prod_{\substack{j > 5n \\ 5 \nmid j}} (1 - q^j)^{-1}, \quad B_n = GT_n. \quad (2.1)$$

Let the Rogers–Ramanujan series be

$$R_t(Q) = \sum_{j \geq 0} \frac{Q^{j^2 + (t-1)j}}{(Q; Q)_j}, \quad g = R_1, \quad h_R = R_2 \quad (2.2)$$

The classical sum-product identities [RR] give

$$g(Q) = \frac{1}{(Q, Q^4; Q^5)_\infty}, \quad h_R(Q) = \frac{1}{(Q^2, Q^3; Q^5)_\infty}.$$

The classical 5-dissection [W21, proof of Proposition 3.6] is

$$G(q) = g(q^5)^2 - qg(q^5)h_R(q^5) - q^2h_R(q^5)^2. \quad (2.3)$$

This form also follows from the continued-fraction representation in the same reference by setting $Q = q^5$, taking the continued-fraction product ratio to be $h_R(Q)/g(Q)$, and multiplying by $(q^{25}; q^{25})_\infty / (q^5; q^5)_\infty = g(Q)h_R(Q)$.

The three series in Q on the right have nonnegative coefficients, and

$$[q^{5j+3}]G = [q^{5j+4}]G = 0. \quad (2.4)$$

Consequently, if $m \equiv 3, 4 \pmod{5}$, then exactly

$$c_n(m) = [q^m]\{G(q)(T_n(q) - 1)\}. \quad (2.5)$$

Equation (2.5) is used in the coefficient integral over the entire circle. It is not an operation that subtracts G only from the main term on the major arc.

Moreover, every coefficient of g^2 and gh_R is positive. In h_R^2 , only the coefficient of Q^1 vanishes; those of Q^0 and Q^K for $K \geq 2$ are positive. The latter follows because every integer $K \geq 2$ can be formed from parts 2 and 3. Hence $[q^7]G = 0$, and we do not invoke an unconditional strict negative sign that would include the small indices.

3. A nonnegative kernel and an exact formula for the first tail coefficients

From the definition, canceling the factor $1 - Q^j$ in the numerator of the difference and writing $j = i + 1$ gives

$$R_t = R_{t+1} + Q^t R_{t+2}. \quad (3.1)$$

Put $W_t = R_{t+1}^2 - QR_t R_{t+2}$. Substitution of (3.1) yields

$$W_t = (1 - Q)R_{t+1}R_{t+2} + Q^{2t+2}W_{t+2}. \quad (3.2)$$

Iterating and sending the lowest degree of the remainder to infinity, as a formal power series we obtain

$$\frac{R_{t+1}^2 - QR_t R_{t+2}}{1 - Q} = \sum_{j \geq 0} Q^{2j(t+j)} R_{t+2j+1} R_{t+2j+2}. \quad (3.3)$$

In particular, since $QR_3 = g - h_R$,

$$\mathcal{D}(Q) := \frac{gh_R + h_R^2 - g^2}{1 - Q} = \sum_{j \geq 0} Q^{2j(j+1)} R_{2j+2} R_{2j+3} = \sum_{K \geq 0} d_K Q^K. \quad (3.4)$$

The right-hand side is nonnegative, with $d_0 = 1, d_1 = 0$. The $j = 0$ term contains $Q^2/(1 - Q)$, so $d_K > 0$ for $K \geq 2$.

Expanding by the number of factors selected from the tail gives

$$T_n(q) = 1 + q^{5n} \frac{q + q^2 + q^3 + q^4}{1 - q^5} + O(q^{10n+2}). \quad (3.5)$$

After multiplication by (2.3), both residue classes 3 and 4 contain $(g^2 - gh_R - h_R^2)/(1 - Q) = -\mathcal{D}$. Therefore

$$\boxed{c_n(5n + 5K + a) = -d_K \quad (a = 3, 4, 0 \leq K \leq n - 1).} \quad (3.6)$$

The relevant degrees are at most $10n - 1$, so the remainder in (3.5) does not contribute. This formula holds for every n and is not extended to $K \geq n$.

4. The endpoint main term and explicit error

Define τ, w_0, x_0 by

$$-R'(\tau) = \frac{k}{5n^2}, \quad w_0 = \frac{\tau}{5n}, \quad x_0 = e^{-\tau},$$

and assume

$$\tau \geq 11/2, \quad 0 < w_0 \leq 0.0013 \quad (4.1)$$

The positive main term and error budget are

$$P(n, k) = \frac{\exp\{nR(\tau) + kw_0 - w_0/6\}}{5\sqrt{2\pi R''(\tau)} n^{3/2}}, \quad (4.2)$$

$$\mathcal{E}(\tau, w_0) = 15\sqrt{w_0} + 15e^{-\tau} + 60w_0 + 50w_0^{-3}e^{-1/(30w_0)}. \quad (4.3)$$

Endpoint Estimate E

Taking $k = m - 5n$ in the two weak residue classes and $k = m$ in the three strong residue classes,

$$\left| \frac{c_n(m)}{-P(n, m - 5n)} - 1 \right| \leq \mathcal{E} \quad (m \equiv 3, 4 \pmod{5}), \quad (4.4)$$

$$\left| \frac{c_n(m)}{\sigma_a P(n, m)} - 1 \right| \leq \mathcal{E} \quad (m \equiv a \pmod{5}, a = 0, 1, 2), \quad (4.5)$$

$$(\sigma_0, \sigma_1, \sigma_2) = \left(\frac{5 + \sqrt{5}}{2}, -\sqrt{5}, -\frac{5 - \sqrt{5}}{2} \right).$$

The error decreases with τ and increases with w_0 . Monotonicity of the last term follows from $1/30 - 3w_0 > 0$. Hence

$$\mathcal{E} \leq \mathcal{E}(11/2, 0.0013) = 0.84662267115518262237 \dots < 0.85 < 1. \quad (4.6)$$

We derive (4.4)–(4.5) in the next section.

5. Proof of the endpoint estimate

5.1. An all-angle gap for positive-coefficient periodic products

Let $\mathcal{P} \in \{g^2, gh_R, h_R^2\}$. The product weights have period 5, and their weights on the nonzero residue classes are, respectively,

$$(2, 0, 0, 2), \quad (1, 1, 1, 1), \quad (0, 2, 2, 0).$$

They satisfy $\sum b_r = 4$, $\sum r b_r = 10$. If $0 < v \leq 0.0065$ and $\text{dist}(\theta, 2\pi\mathbb{Z}) \geq 3v/4$, then

$$|\mathcal{P}(e^{-v+i\theta})| \leq \mathcal{P}(e^{-v})e^{-1/(5v)}. \quad (5.1)$$

Indeed, retaining only the first term of the logarithmic series gives a loss of at least

$$S = \sum_{k \geq 1} b_k e^{-vk} (1 - \cos k\theta) = \frac{N_v(0)}{1 - e^{-5v}} - \Re \frac{N_v(\theta)}{1 - e^{-5v+i5\theta}}, \quad N_v(\theta) = \sum_{r=1}^4 b_r e^{-rv+i r\theta}.$$

Write $\theta = 2\pi b/5 + \epsilon$, $|\epsilon| \leq \pi/5$. When $|\epsilon| \geq 2v$, the squared denominator and $1 - \cos t \geq 2t^2/\pi^2$ for $|t| \leq \pi$ give

$$S \geq \frac{4 - 10v}{5v} \left(1 - \frac{1}{\sqrt{1 + 16e^{-5v}/\pi^2}} \right) \geq \frac{0.2}{v}.$$

When $|\epsilon| \leq 2v$, put $t = \epsilon/v$ and $C_b = \sum b_r e^{2\pi i b r/5}$. Because the weights are symmetric, C_b is real; $C_0 = 4$, while $C_b \leq \sqrt{5} - 1 < 1.25$ if $b \not\equiv 0 \pmod{5}$.

$$S = \frac{4 - C_b/(1 + t^2)}{5v} + E, \quad |E| < 13. \quad (5.2)$$

Here we use $|N_v(\theta) - C_b| \leq 30v$ and $|1 - e^{-5v+i5\epsilon}| \geq 5v/(1 + 5v)$. For $d = 5v - i5\epsilon$, $|d| \leq 5\sqrt{5}v$, [Coth] gives

$$\left| \frac{1}{1 - e^{-d}} - \frac{1}{d} \right| \leq \frac{1}{2} + \frac{|d|}{12(1 - |d|^2/(4\pi^2))} < 0.507.$$

The numerator- and denominator-replacement errors on the nonzero-angle side are at most $6.195 + 2.028$, and the error on the real-axis side is at most $2 + 4(0.507) = 4.028$, for a total

below 13. If $b = 0$, then $|t| \geq 3/4$, so the main term is at least $0.288/v$; if $b \neq 0$, it is at least $0.55/v$. After subtracting $13v \leq 0.0845$, it is still at least $0.2/v$, proving (5.1).

This argument is not a check obtained by sampling angles. Scalar comparisons are handled by certificates E01–E06.

5.2. The eta transformation and an upper bound on the positive real axis

Applying the transformation law [Eta] to $q = \zeta^\ell e^{-w}$ gives

$$G(\zeta^\ell e^{-w}) = \kappa_\ell e^{A/w-w/6} U_\ell(w), \quad (\kappa_1, \kappa_2, \kappa_3, \kappa_4) = (e^{-i\pi/5}, 1, 1, e^{i\pi/5}), \quad (5.3)$$

$$U_\ell(w) = \frac{(\zeta^{-\ell^{-1}} e^{-4\pi^2/(25w)}; \zeta^{-\ell^{-1}} e^{-4\pi^2/(25w)})_\infty}{(e^{-4\pi^2/(5w)}; e^{-4\pi^2/(5w)})_\infty}. \quad (5.4)$$

Here ℓ^{-1} is the inverse modulo 5. The multiplier uses the Dedekind sums $s(\ell, 5) = (1/5, 0, 0, -1/5)$. Also,

$$G(e^{-w}) = \sqrt{5} e^{-C/w-w/6} U_0(w), \quad U_0(w) = \frac{(e^{-4\pi^2/w}; e^{-4\pi^2/w})_\infty}{(e^{-4\pi^2/(5w)}; e^{-4\pi^2/(5w)})_\infty}. \quad (5.5)$$

For $w = w_0 + iy$, $|y| \leq 3w_0/4$, we have $\Re(1/w) \geq 16/(25w_0)$. From $|\log(z; z)_\infty| \leq |z|/(1 - |z|)^2$ and $(4\pi^2/25)(16/25) > 1$, throughout the range (4.1),

$$|U_\ell(w) - 1| \leq 10e^{-1/w_0} \quad (\ell = 0, \dots, 4). \quad (5.6)$$

Use the root filter in (2.3) to extract the three $\mathcal{P}$. With $w = v/5$, each primitive-root term is at most $1.001e^{A/w}$ and the $q = 1$ term at most $0.001e^{A/w}$. Removing the coefficients q, q^2 costs a factor at most e^{2w} , so

$$\mathcal{P}(e^{-v}) \leq 2e^{C/v} \quad (0 < v \leq 0.0065). \quad (5.7)$$

Certificates E07–E10 and E36–E37 verify the numerical margins.

5.3. Tail expansion and combination of the four arcs

With $x = e^{-5nw}$ and $|x| = x_0$, put

$$\Lambda(x) = \frac{1}{5} \text{Li}_2(x^5) - \text{Li}_2(x), \quad L_\ell(x) = \sum_{\substack{j \geq 1 \\ 5 \nmid j}} \frac{x^j}{j} \left(\frac{\zeta^{\ell j}}{1 - \zeta^{\ell j}} + \frac{1}{2} \right).$$

The exact tail logarithm is

$$\log T_n(\zeta^\ell e^{-w}) = \sum_{j \geq 1} \frac{x^j}{j} \left\{ \frac{1}{\zeta^{-\ell j} e^{jw} - 1} - \frac{1}{e^{5jw} - 1} \right\}.$$

Expanding it gives

$$\log T_n(\zeta^\ell e^{-w}) = \frac{\Lambda(x)}{5w} + L_\ell(x) + E_\ell(w, x), \quad |E_\ell(w, x)| \leq \frac{5|w|x_0}{1-x_0}. \quad (5.8)$$

We verify uniformity of the error. If $|\Im t| \leq 3\Re t/4$, then for a nontrivial fifth root ξ ,

$$\left| \frac{d}{dt} \frac{1}{\xi^{-1} e^t - 1} \right| = \frac{1}{2(\cosh \Re t - \cos(\Im t - \arg \xi))} \leq 4.$$

For $\Re t \leq 1$ use the angular distance $2\pi/5 - 3/4$, and for $\Re t \geq 1$ use $\cosh 1 - 1$. Also, from $\Re(t^2) \geq 0$ and the partial-fraction expansion of $\coth$,

$$\left| \frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} \right| \leq \frac{|t|}{12}.$$

Thus the first-order remainder in each brace is less than $5j|w|$ when $5 \nmid j$, and at most $j|w|/2$ when $5 \mid j$. Summing the series gives (5.8). We have not assumed that $x_0/w_0 \rightarrow 0$.

If $C_a = \sum_{\ell=1}^4 \zeta^{-a\ell} \kappa_\ell$, then

$$(C_0, C_1, C_2, C_3, C_4) = (\sigma_0, \sigma_1, \sigma_2, 0, 0).$$

For the two weak residue classes, $S_a(x) = \sum \zeta^{-a\ell} \kappa_\ell e^{L_\ell(x)}$ satisfies $S_a(0) = 0, S'_a(0) = -1$. The derivative calculation is checked as follows. If $V_a = \sum \zeta^{-a\ell} \kappa_\ell / (1 - \zeta^\ell)$, then $V_a - V_{a-1} = C_a$ and $\sum_a V_a = 0$. Since $C_3 = C_4 = 0$, we obtain $V_2 = V_3 = V_4 = -1$, and hence $S'_a(0) = V_{a-1} + C_a/2 = -1$.

The absolute value of the coefficient of degree j in L_ℓ is at most $1/j$, so

$$|S_a(x) + x| \leq 8x_0^2.$$

The error in the four-root sum resulting from exponentiating (5.8) is at most $25|w|x_0$. Using $(C + \Lambda(x))/(5w) = nR(5nw)$ gives

$$\sum_{\ell=1}^4 \zeta^{-a\ell} \kappa_\ell e^{A/w-w/6} (T_n(\zeta^\ell e^{-w}) - 1) = -x e^{nR(5nw)-w/6} (1 + \epsilon(w)), \quad (5.9)$$

$$|\epsilon(w)| \leq 8x_0 + 25|w|.$$

The term -1 cancels exactly because $C_a = 0$. The eta remainder is always retained in the form $(G - G_{\text{main}})(T_n - 1)$. For the three strong residue classes, use $S_a(0) = \sigma_a \neq 0$; the same relative-error bound applies. For example, $|S_a(x) - \sigma_a|/|\sigma_a| \leq 8x_0$ follows from $\min |\sigma_a| = (5 - \sqrt{5})/2 > 1$. The numerical comparisons are E11–E15 and E35.

5.4. Integrals over the minor arcs and of the eta remainder

Let $r = e^{-w_0}$. Positivity of the tail coefficients gives, on the entire circle,

$$|T_n(q) - 1| \leq T_n(r) - 1 \leq \frac{x_0}{w_0} e^{x_0/w_0}. \quad (5.10)$$

Indeed, estimating the tail logarithm by geometric series gives

$$\log T_n(r) \leq 4 \frac{x_0}{1-x_0} + \frac{4}{5w_0} \frac{x_0}{1-x_0} \leq \frac{x_0}{w_0}.$$

On the region whose angular distance from all five fifth roots is at least $3w_0/4$, applying (5.1) and (5.7) to $Q = q^5$ gives

$$|G(q)| \leq 6 \exp \left(\frac{A}{w_0} - \frac{0.04}{w_0} \right). \quad (5.11)$$

In the weak case retain the factor (5.10), so $x_0 r^{-m} = r^{-k}$, where $k = m - 5n$. From $\Lambda(x_0) \geq -x_0/(1-x_0)$ and

$$0.04 - x_0 - \frac{x_0}{5(1-x_0)} > \frac{1}{30},$$

the minor-arc contribution divided by (4.2) is at most

$$16w_0^{-3} e^{-1/(30w_0)}.$$

Specifically, §5.5 gives $E''(w_0) \leq 0.67/w_0^3$, so the prefactor is $6\sqrt{2\pi \cdot 0.67} e^{w_0/6} w_0^{-5/2}$, bounded by $16w_0^{-3}$.

Near $q = 1$ use (5.5), and for the eta remainder at the primitive roots use (5.6). Each exponential gap from the main term is at least $0.9/w_0$; even the crude prefactor $100w_0^{-5/2}$ is absorbed into $w_0^{-3} e^{-1/(30w_0)}$. With margin, allocate to the entire excluded part

$$50w_0^{-3} e^{-1/(30w_0)} \quad (5.12)$$

In the strong case use $T_n(r) \leq e^{x_0/w_0}$; with $k = m$ the same bound results. The comparisons are E16–E21.

5.5. Gaussian replacement and the prefactor

$$F(w) = nR(5nw) = \frac{A}{w} + Q_n(w), \quad Q_n(w) = \frac{1}{5w} \sum_{j \geq 1} \frac{a_j e^{-5njw}}{j^2},$$

where $a_j = -1$ if $5 \nmid j$, and $a_j = 4$ if $5 \mid j$. For $r = 2, 3$ and $|y| \leq 3w_0/4$,

$$w_0^{r+1} |Q_n^{(r)}(w_0 + iy)| \leq \frac{4}{5} r! \sum_{j \geq 1} \frac{x_0^j}{j^2} \sum_{h=0}^r \frac{(\tau j)^h}{h!}. \quad (5.13)$$

For $\tau \geq 11/2$, $h \leq 3$, $\tau^h e^{-\tau j}$ decreases with τ . Set $T_* = 11/2$, $x_* = e^{-T_*}$, and

$$B_2 = \frac{8}{5} (1 + T_* + T_*^2/2) \frac{x_*}{1 - x_*} < 0.142,$$

$$B_3 = \frac{24}{5} \left[(1 + T_* + T_*^2/2) \frac{x_*}{1 - x_*} + \frac{T_*^3}{6} \frac{x_*}{(1 - x_*)^2} \right].$$

Writing $E(w) = F(w) + kw$, the saddle condition $E'(w_0) = 0$ and

$$\frac{0.38}{w_0^3} \leq E''(w_0) \leq \frac{0.67}{w_0^3}, \quad |E'''(w_0 + iy)| \leq \frac{2.56}{w_0^4} \quad (5.14)$$

follow. Computing the real part of A/w exactly and bounding the quadratic term of Q_n gives

$$\Re E(w_0 + iy) - E(w_0) \leq -\frac{0.097y^2}{w_0^3}, \quad (5.15)$$

because $A/(1 + (3/4)^2) - B_2/2 > 0.097$.

Linear interpolation between the two exponents gives

$$\left| e^{E(w_0 + iy) - E(w_0)} - e^{-E''(w_0)y^2/2} \right| \leq \frac{2.56|y|^3}{6w_0^4} e^{-0.097y^2/w_0^3}.$$

Integrating and dividing by $J_0 = \sqrt{2\pi/E''(w_0)}$ gives at most $14.9\sqrt{w_0}$. Including the complete Gaussian tail gives at most $15\sqrt{w_0}$.

The error from the amplitude in (5.9) and from $e^{-iy/6}$ is at most $8x_0 + 32w_0$. Specifically, $|w| \leq 5w_0/4$ and $|e^{-iy/6} - 1| \leq (w_0/8)e^{w_0/8}$ show that the coefficient of w_0 is

$$31.25 + e^{w_0/8}(1 + 8x_0 + 31.25w_0)/8 < 32.$$

The comparison remains valid under the new upper bounds by substituting $x_0 < 0.005$, $w_0 < 0.002$. The ratio of the absolute integral to J_0 is at most $\sqrt{0.67/0.194} < 1.86$, so the amplitude contributes at most $15x_0 + 60w_0$. Together with (5.12), this gives (4.3). The comparisons are E22–E33.

After combining the four arcs with the same y , the weak-case factor $-xe^{mw} = -e^{kw}$ shifts the location by $5n$. No extra “factor 4 from the four roots” is multiplied in at this stage.

$$\frac{e^{E(w_0) - w_0/6}}{2\pi} \sqrt{\frac{2\pi}{E''(w_0)}} = \frac{e^{nR(\tau) + kw_0 - w_0/6}}{5\sqrt{2\pi R''(\tau)} n^{3/2}},$$

where $E''(w_0) = 25n^3 R''(\tau)$ was used. This is (4.2), and (4.4)–(4.5) follow.

6. The endpoint region for $n \geq 31147$

$$\rho = -5R'(11/2) = 0.2131128927124522702188745809\dots \quad (6.1)$$

Since $\Lambda(e^{-\tau}) < 0$ and

$$\frac{d}{d\tau}\Lambda(e^{-\tau}) = \log((1 - e^{-5\tau})/(1 - e^{-\tau})) > 0,$$

we have

$$0 < -R'(\tau) < C/\tau^2, \quad w_0^2 < C/(5k). \quad (6.2)$$

If $k \geq 5n$ and $n \geq 31147$, then

$$w_0^2 < C/(25n) \leq (0.0013)^2.$$

Comparison E34 verifies $C/(25(0.0013)^2) < 31147$.

Consider $0 \leq m \leq \rho n^2$.

- For $a = 0, 1, 2$: when $m \leq 5n$, the coefficient agrees with that of G ; when $m > 5n$, apply the endpoint estimate with $k = m$.
- For $a = 3, 4$: when $m \leq 5n$, it is zero; when $0 < k = m - 5n < 5n$, use (3.6); when $k \geq 5n$, apply the endpoint estimate.

Whenever a saddle is needed, $k \leq m \leq \rho n^2$, so $\tau \geq 11/2$. Thus all coefficients at the left endpoint have the desired sign. The right endpoint follows from (1.3). The zero coefficients in this range occur only at the positions listed in (1.2).

The finite decimal expansion of ρ must not be rounded upward to enlarge the domain. The region is defined by the exact function value in (6.1).

7. Sign and quantitative lower bound for the interior main phase

Define the amplitudes at the primitive roots by

$$\lambda_\ell(z) = \sum_{j=1}^4 \left(\frac{j}{5} - \frac{1}{2} \right) \{ \log(1 - \zeta^{\ell j} e^{-z}) - \log(1 - \zeta^{\ell j}) \}, \quad \mathcal{A}_\ell(z) = e^{\lambda_\ell(z)}.$$

The branches are chosen continuously from $z = 0$. On the real axis, pairing the conjugate terms j and $5 - j$ gives $|\mathcal{A}_\ell(\tau)| = 1$.

$$\Phi_a(e^{-\tau}) = \sum_{\ell=1}^4 \zeta^{-a\ell} \mathcal{A}_\ell(\tau). \quad (7.1)$$

Put $x = e^{-\tau}$, $v = (1 - x)/(1 + x)$, and

$$\alpha = \arctan(v \cot(\pi/5)), \quad \beta = \arctan(v \cot(2\pi/5)), \quad U = (\alpha + 2\beta)/5, \quad V_0 = (2\alpha - \beta)/5.$$

Pairing conjugate roots and converting a sum of two cosines into a product gives

$$\Phi_a(x) = 4 \cos(3\pi a/5 + U) \cos(-\pi a/5 + V_0). \quad (7.2)$$

Both U and V_0 increase strictly from 0 to $\pi/10$ as v ranges over $[0, 1]$. For example,

$$V'_0 \geq v_* = \frac{\sin(2\pi/5) - \cot(2\pi/5)}{5} > 0, \quad U' \geq u_* = \frac{\sin(\pi/5) \cos(\pi/5) + 2 \sin(2\pi/5) \cos(2\pi/5)}{5} > 0.$$

The ranges of the angles imply $\Phi_0 > 0$ and $\Phi_a < 0$ for $a \neq 0$ and $x > 0$. For the two weak residue classes,

$$-\Phi_3 = 4 \cos(\pi/5 - U) \sin(\pi/10 - V_0), \quad -\Phi_4 = 4 \sin(\pi/10 - U) \cos(\pi/5 + V_0).$$

Using $\sin y \geq 2y/\pi$ and $1 - v \geq x$, we obtain $8 \cos(\pi/5)v_*/\pi > 1/4$ and $8 \cos(3\pi/10)u_*/\pi > 1/4$. The three strong residue classes have a positive constant lower bound. Hence

$$\epsilon_a \Phi_a(x) \geq x/4, \quad \epsilon_0 = 1, \quad \epsilon_1 = \dots = \epsilon_4 = -1. \quad (7.3)$$

The numerical comparisons are B01–B05.

8. All-angle localization with denominator-dependent parameters

Let $n \geq N = 31147$, $0 \leq \tau \leq 11/2$, and $r = e^{-\tau/(5n)}$. In Dirichlet approximation we use the common value $Q = 4139$ throughout and choose a reduced fraction a/b , with $b \leq Q$, such that $|\theta/(2\pi) - a/b| \leq 1/(bQ)$. Put $t = 5n(\theta - 2\pi a/b)$. The following parameters are used only after the approximant has been chosen, so they do not alter the covering of the circle.

denominator	η	K	L	upper bound κ_* for the geometric-series kernel
$5 \nmid b$	11313/1250000	718	$\log(2/\eta)$	$\kappa(Z_1)$
$5 \mid b$	113/40000	2717	$\log(2/\eta)$	$\kappa(Z_5)$

Throughout this section, η, K, L mean the values in the applicable row. Set $Z_1 = (11/2)K/N + 10\pi K/Q$ and $Z_5 = (11/2)K/N + 2\pi K/Q$, and use the function κ of §8.3 directly in interval arithmetic. No rounded fixed constant is substituted into the error budget. The truncation tail is $\delta = 10e^{-\eta K}/(\eta K)$, and the error in the resonant sums is $4\kappa_* L/n$.

8.1. Competition with smoothed denominators not divisible by 5

Let $h(z) = \int_0^1 \log(1 - e^{-zx}) dx$ and $M(\tau) = \sup_t \Re h(\tau - it)$. The maximum principle in the right half-plane, together with $\Re h \rightarrow -\infty$ at the origin and $h \rightarrow 0$ at infinity, shows that M is nonnegative and nonincreasing. Put $f(y) = \log |2 \sin(y/2)|$. On the unit circle the mean is $t^{-1} \int_0^t f(y) dy$. Each subsequent period moves the mean of the first period toward zero. The unique zero of $H(t) = tf(t) + \sum_{k \geq 1} \sin(kt)/k^2$ in $(\pi, 2\pi)$ is the maximum point. Bounding 1,000 terms and the tail by $1/1000$ encloses $H(4.97) > 0$ and $f(4.97) < 1/5$, and therefore $M(0) < 1/5$.

The numerator radius is retained in the smoothing increment. If $0 \leq \rho \leq 1$, then

$$|1 - \rho e^{iv}| \geq \rho |1 - e^{iv}|, \quad \frac{|1 - \rho e^{-\delta+iv}|}{|1 - \rho e^{iv}|} \leq 1 + \frac{\rho(1 - e^{-\delta})}{|1 - \rho e^{iv}|} \leq 1 + \frac{\delta}{2|\sin(v/2)|}. \quad (8.19)$$

The difference of the squares in the first inequality is $(1 - \rho)(1 + \rho - 2\rho \cos v) \geq 0$. The singularities have measure zero, and the logarithmic upper bound is locally integrable. If $\mathcal{L}(u) = (1 + u) \log(1 + u) - u \log u$, integration using $|\sin(v/2)| \geq |v|/\pi$ for $|v| \leq \pi$ shows that the mean over a period is at most $\mathcal{L}(\delta/2)$. The mean over an interval from the origin of length $s \geq 1$ is at most 2π times the periodic mean; if $s \geq 2$, it is at most π times that mean. For an interval shorter than one period, bound it by the entire period; for a longer interval, split it into complete periods and a remainder. The concavity of $\mathcal{L}$ implies that $\mathcal{L}(bu)/b$ is nonincreasing for $b \geq 1$.

Another bound that retains the real radius is, for $u > 0$,

$$\log |1 - e^{-u-\delta+iv}| - \log |1 - e^{-u+iv}| \leq \log(1 + \delta/u). \quad (8.16)$$

This follows by integrating the derivative bound $1/(e^u - 1) \leq 1/u$ for variation of the radius. With $u = b\tau x$ and $\delta = b\eta$, integration bounds the increment by $\mathcal{L}(\eta/\tau)$.

Write $\mathcal{H}_b$ for the resonant main term below. First suppose $\tau \leq 1/2$. Then $R(\tau) \geq R(1/2) > 1.189$. For $b = 1$ and $|t| \leq 2$, the inequality $|1 - \rho e^{itx}| \leq \max(1, 2|\sin(tx/2)|)$, valid for every radius, gives

$$\mathcal{H}_1 \leq 4(1 - \pi/6) \log(2 \sin 1).$$

Indeed, the logarithm is nonpositive when $|tx| \leq \pi/3$, and the remaining interval has length at most $1 - \pi/6$. If $|t| \geq 2$, then $\mathcal{H}_1 \leq 0.8 + 4\pi\mathcal{L}(\eta/2)$. In either case, $R - \mathcal{H}_1 > 0.025$. For $b \geq 2$, every factor has absolute value at most 1 when the angular length $b|t| < 1$; otherwise $\mathcal{H}_b \leq 0.4 + 4\pi\mathcal{L}(\eta)$, so $R - \mathcal{H}_b > 0.1$.

For $\tau \geq 1/2$, put $A_0 = \pi^2/6$ and $B(\tau) = \text{Li}_2(e^{-\tau})$. Then $M(\tau) \leq B(\tau)^2/(4A_0\tau)$ and $R(\tau) \geq (C - B(\tau))/\tau$. The first inequality follows from $|\text{Li}_2(e^{-\tau+it})| \leq B(\tau)$ and maximization of a quadratic. Cover $[1/2, 11/2]$ by 200 intervals $[u, v]$ of width $1/40$ and enclose

$$\frac{C - B(u) - B(u)^2/A_0}{v} - 4\mathcal{L}(\eta/u) > 0.025, \quad \frac{C - B(u) - B(u)^2/(2A_0)}{v} - 2\mathcal{L}(\eta/u) > 0.1 \quad (8.20)$$

The first inequality applies to $b = 1$, and the second to $b \geq 2$. The upper bound for B uses 100 terms and a positive geometric-series tail. The proof of $R(1/2) > 1.189$ also uses five positive terms of $\text{Li}_2(e^{-5/2})$. Throughout the range, $R > 0.238$. These statements are checked by L00a–L11.

8.2. Nonresonant error frequency by frequency

The inequality $|1 - ue^{iv}| \leq e^{\eta/2} |1 - ue^{-\eta+iv}|$ gives

$$\frac{1}{n} \log |B_n(re^{i\theta})| \leq 2\eta - \frac{1}{n} \Re \sum_{l \geq 1} \frac{e^{-\eta l}}{l} \left(\sum_{j=1}^{5n} e^{-l\tau j/(5n)+ilj\theta} - \sum_{j=1}^n e^{-l\tau j/n+i5lj\theta} \right). \quad (8.1)$$

For $d = 1, 5$, put $g = \gcd(b, d)$ and $B = b/g$. The following nonresonant sums are always taken over the subset satisfying $b \nmid dl$. If $r_{d,l} = \text{dist}(dla, b\mathbb{Z})$, then

$$\left\| \frac{dl\theta}{2\pi} \right\| \geq \frac{r_{d,l} - dl/Q}{b}, \quad \left| \sum e^{-\alpha j + idlj\theta} \right| \leq \frac{b}{2(r_{d,l} - dl/Q)}. \quad (8.17)$$

The latter follows from the geometric series and Abel summation for monotone positive weights. Writing the distance as $r_{d,l} = gr$, and assuming $\epsilon = dK/(gQ) < 1$, we have

$$\frac{1}{r - dl/(gQ)} \leq \frac{1}{r} + \frac{dl}{gQ(1 - \epsilon)r^2}. \quad (8.21)$$

Within each block of length B , each nonzero distance r occurs at most twice. When the decreasing positive weights are rearranged, missing resonant terms may be filled with zeros without invalidating the upper bound.

$$S_0 = \sum_{l=1}^K \frac{e^{-\eta l}}{l \lceil l/2 \rceil}, \quad T(B) = S_0 + \frac{2(1 + \log K)e^{-\eta B}}{B(1 - e^{-\eta B})}, \quad U(B) = \frac{\pi^2}{3(1 - e^{-\eta B})}. \quad (8.22)$$

The first block is at most S_0 . The remainder of the j th block, for $j \geq 1$, is bounded using $e^{-\eta jB}/(jB)$ and $\sum_{i \leq K} 1/\lceil i/2 \rceil \leq 2(1 + \log K)$. The inverse-square sum in one block is at most $2 \sum_{r \geq 1} r^{-2} = \pi^2/3$. Consequently, the normalized nonresonant error on the d side is at most

$$\frac{b}{2gn} \left[T(B) + \frac{d}{gQ(1 - \epsilon)} U(B) \right] \quad (8.23)$$

Both T and U decrease as B increases. If $5 \nmid b$ and $b > 2K$, two frequencies at the same distance would satisfy $l_1 \equiv \pm l_2 \pmod{b}$. For two distinct frequencies this would require $l_1 + l_2 = b > 2K$, which is impossible. In this case each distance occurs at most once, so T, U in (8.23) may be replaced respectively by $S_1 = \sum_{l \leq K} e^{-\eta l}/l^2$ and $\pi^2/6$. This removes both the loss incurred by applying dK/Q to every frequency and the loss from double occurrence for large denominators.

8.3. Resonant sums and classification of all denominators

For $A = l(\tau - it)$ and $M = n$ or $5n$,

$$\sum_{j=1}^M e^{-Aj/M} - M \int_0^1 e^{-Ax} dx = (1 - e^{-A}) \left(\frac{1}{e^{A/M} - 1} - \frac{M}{A} \right). \quad (8.12)$$

At $A = 0$ this is interpreted by continuous extension. The partial-fraction expansion of $\coth$ gives, for $|z| < 2\pi$,

$$\left| \frac{1}{e^z - 1} - \frac{1}{z} \right| \leq \kappa(|z|) = \frac{1}{2} + \frac{|z|}{12(1 - |z|^2/(4\pi^2))}. \quad (8.13)$$

Here $|A/M| \leq (11/2)K/N + 10\pi K/(bQ)$. Using $b \geq 1$ in the case $5 \nmid b$ and $b \geq 5$ in the case $5 \mid b$ gives the tabulated value of κ_* . Since $|1 - e^{-A}| \leq 2$, the total error from the two resonant sums is at most $4\kappa_*L/n$. The tail coefficients for the finite sums and continuous main terms are at most 4 and 6 respectively, so their combined tail is at most δ . The resonant main term is

$$\mathcal{H}_b = \begin{cases} \frac{4}{b} \int_0^1 \log |1 - e^{-b\eta - b(\tau - it)x}| dx, & 5 \nmid b, \\ \frac{1}{j} \int_0^1 \log \left| \sum_{r=0}^4 e^{-rj(\eta + (\tau - it)x)} \right| dx, & b = 5j. \end{cases} \quad (8.3)$$

For $b = 5j$ and $j \geq 2$, $\mathcal{H}_b \leq R(j\tau)/j \leq R(\tau)/2$. In general, $4 \log 2/b$ is an upper bound when $5 \nmid b$, and $5 \log 5/b$ when $5 \mid b$. The coefficient γ_b in the rough nonresonant bound $\gamma_b bL/n$ is $[2(1 - K/Q)]^{-1} + [2(1 - 5K/Q)]^{-1}$ for denominators not divisible by 5, and $0.6/(1 - K/Q)$ for denominators divisible by 5. When $b = 5$, all terms on the $d = 5$ side are resonant, so $\gamma_5 = [2(1 - K/Q)]^{-1}$. When $b = 1$, there are no nonresonant terms.

denominator	gap from the main term, or upper bound for the main term	nonresonant estimate
$b = 1$	gap 0.025	0
$2 \leq b \leq 100$, $5 \nmid b$	gap 0.1	rough bound, $b \leq 100$
$100 < b \leq 1436$, $5 \nmid b$	main term at most $4 \log 2/101$	(8.23), $b \leq 1436, B \geq 101$
$1436 < b \leq Q$, $5 \nmid b$	main term at most $4 \log 2/1437$	single occurrence: $S_1, \pi^2/6$
$5 < b \leq 300$, $5 \mid b$	gap $0.238/2$	rough bound, $b \leq 300$
$300 < b \leq Q$, $5 \mid b$	main term at most $5 \log 5/301$	(8.23), $B_1 \geq 301, B_5 \geq 61$
$b = 5$, outside the major arcs	next-term gap 0.00992	$\gamma_5 5L/n$

In each row, a gap greater than $1/1000$ remains after subtracting 2η , the resonant error, the nonresonant error, and the tail. The comparison is made at the threshold N ; thereafter the $1/n$ terms and the argument bounds for the geometric series decrease. All interval comparisons are L12–L19 in [localization_certificate.json](#).

8.3a. The denominator-5 gap obtained by combining equal differences

Let $p_j(z) = e^{-jz} / \sum_{r=0}^4 e^{-rz}$ and $z = \eta + \tau x$. Before replacing individual products by their worst-case values, combine them for each difference d :

$$W_d(z) = \sum_{j=0}^{4-d} p_j(z) p_{j+d}(z) = \frac{\sum_{j=0}^{4-d} y^{2j+d}}{(1 + y + y^2 + y^3 + y^4)^2}, \quad y = e^{-z} \quad (8.24)$$

Writing $S(y) = 1 + y + \dots + y^4$, we have $dW_d/dy = (1 - y)Q_d(y)/S(y)^3$. In ascending order, the coefficients are

d	coefficients of Q_d
1	(1, 0, 0, -4, -7, -7, -4, 0, 0, 1)
2	(0, 2, 2, 4, 2, 2, 4, 2, 2)
3	(0, 0, 3, 4, 8, 8, 4, 3)
4	(0, 0, 0, 4, 6, 6, 4)

For $d = 2, 3, 4$, the function is increasing in y and hence decreasing in z . For $d = 1$, the coefficients of $(1+t)^9 Q_1(t/(1+t))$ are (1, 9, 36, 80, 95, 24, -98, -148, -90, -20). There is one sign change, so Descartes' rule of signs, together with $Q_1(0) > 0$ and $Q_1(1) < 0$, shows that there is exactly one zero in $(0, 1)$. Thus $W_1(z)$ is unimodal and its minimum on an interval occurs at an endpoint. These polynomial identities are also checked by integer arithmetic.

Let $m = 400$, set $g_{d,i} = \min(W_d(0), W_d(\eta + (11/2)i/m))$, and put $g_{d,m+1} = 0$. This is a monotone lower envelope. For each characteristic function,

$$-\log \left| \sum_j p_j e^{ijtx} \right| \geq \sum_{d=1}^4 W_d(z)(1 - \cos(dtx)).$$

With $b_i = i/m$, the integral of the step function is at least $\sum_i (g_{d,i} - g_{d,i+1}) b_i (1 - \text{sinc}(dtb_i))$. For $0 \leq a \leq 2$ and $|v| \geq a$, use

$$1 - \text{sinc } v \geq a^2/6 - a^4/120 \quad (8.8)$$

On the first half-period this follows from monotonicity of sinc and the Taylor lower bound; for $|v| \geq \pi$, it follows from $1 - 1/\pi > 8/15$. If $|t| \geq 6/5$ and $a_{d,i} = \min(2, (6/5)db_i)$, then

$$\Gamma = \sum_{d=1}^4 \sum_{i=1}^{400} (g_{d,i} - g_{d,i+1}) b_i \left(a_{d,i}^2/6 - a_{d,i}^4/120 \right) > 0.00992. \quad (8.9)$$

The interval lower endpoint is greater than 0.00992; all digits are recorded in the certificate. Radial smoothing does not increase the value on the real axis of a sum with positive coefficients, so $\mathcal{H}_5 \leq R - \Gamma$. Taking the major arcs to be $|\theta - 2\pi\ell/5| \leq (6/5)/(5n)$ also handles the case $b = 5$ outside those arcs. Therefore, everywhere outside the major arcs,

$$\frac{1}{n} \log |B_n(re^{i\theta})| \leq R(\tau) - 1/1000. \quad (8.4)$$

The entire angular range has been covered analytically. No extrapolation from a finite sample of angles is used.

9. Combining the four roots before estimating the interior integral

Let $0 \leq \tau \leq 11/2$ and $z = \tau - it$. The small box is $|t| \leq h = 2/5$ and the large box is $|t| \leq 6/5$. For a nontrivial fifth root ξ and $0 \leq x \leq 1$, the quantity $|1 - \xi e^{-zx}|$ exceeds $d = 0.75$ in the small box and 0.05 in the large box. Use the angular distances $2\pi/5 - h$ and $2\pi/5 - 6/5$. Abbreviate $A_\ell = \mathcal{A}_\ell = e^{\lambda_\ell}$.

9.1. First Euler–Maclaurin correction and second-order remainder

For $f \in C^2[0, 1]$,

$$\left| \sum_{r=0}^{n-1} f((r+\alpha)/n) - n \int_0^1 f - B_1(\alpha)(f(1) - f(0)) \right| \leq \frac{1}{8n} \int_0^1 |f''|. \quad (9.16)$$

The Peano kernel on one small interval is $(\alpha - u)_+ - (1 - u)^2/2 - B_1(\alpha)(1 - u)$, whose absolute value is at most $1/8$. Rescale each interval and telescope the endpoint differences. For C^3 functions, also extracting $B_2(\alpha)\Delta f'/(2n)$ gives

$$\sum_{r=0}^{n-1} f((r+\alpha)/n) = n \int_0^1 f + B_1(\alpha)\Delta f + \frac{B_2(\alpha)}{2n}\Delta f' + \varepsilon, \quad |\varepsilon| \leq \frac{1}{n^2} \int_0^1 |f'''|. \quad (9.24)$$

The third-order kernel is $(\alpha - u)_+^2/2 - (1 - u)^3/6 - B_1(\alpha)(1 - u)^2/2 - B_2(\alpha)(1 - u)/2$. Since $|B_1| \leq 1/2$ and $|B_2| \leq 1/6$, its absolute value is at most $1/2 + 1/6 + 1/4 + 1/12 = 1$. The kernel annihilates polynomials of degree at most two. Summing over the small intervals proves (9.24).

Apply this to $f_j(x) = \log(1 - \zeta^{\ell j} e^{-zx})$ with $\alpha = j/5$. The main integral is $nR(z)$, and the first endpoint difference is $\lambda_\ell(z)$ from §7. Set $b_j = B_2(j/5)$. Then

$$B_n(\zeta^\ell e^{-z/(5n)}) = e^{nR(z)} A_\ell(z) e^{D_\ell(z)/n + \varepsilon_{\ell,n}(z)}, \quad |\varepsilon_{\ell,n}| \leq 3400/n^2, \quad (9.25)$$

$$D_\ell(z) = \frac{z}{2} \sum_{j=1}^4 b_j \left(\frac{\zeta^{\ell j} e^{-z}}{1 - \zeta^{\ell j} e^{-z}} - \frac{\zeta^{\ell j}}{1 - \zeta^{\ell j}} \right).$$

In the small box, $|z| < 5.6$ and $|f_j'''(x)| \leq 2|z|^3 e^{-\tau x}/d^3$, so the sum of the four integrals is at most $8(5.6)^3/d^3 < 3400$. In the large box use only (9.16), giving a logarithmic remainder at most $7200/n$.

9.2. Combination retaining the small factor

Put $x = e^{-\tau}$ and $M = 1.54$. On the real axis, $|A_\ell(\tau)| = 1$. In the small box, $|\lambda'_\ell| \leq 0.8x/d$ implies $|A_\ell(z)| \leq e^{0.4(0.8)/d} < M$. In the large box, estimate the amplitude directly from its product representation. The sums of the absolute positive and negative exponents are each 0.4 ; using $1 < |1 - \xi| \leq 2$ and $0.05 < |1 - \xi e^{-z}| \leq 2$ gives $|A_\ell(z)| \leq 80^{2/5} < 6$.

$$a_1 = \frac{0.8Mx}{d}, \quad a_2 = \frac{M}{2} \left(\frac{0.8x}{d^2} + \left(\frac{0.8x}{d} \right)^2 \right). \quad (9.18)$$

Thus the change in $A_\ell(\tau - it)$ from the real axis is at most $a_1|t|$, and the remainder after its linear term is at most $a_2 t^2$.

$$\Psi_a(z) = \sum_{\ell=1}^4 \zeta^{-a\ell} A_\ell(z), \quad \Theta_a(z) = \sum_{\ell=1}^4 \zeta^{-a\ell} A_\ell(z) D_\ell(z). \quad (9.26)$$

We have $\Psi_a(\tau) = \Phi_a(e^{-\tau})$. For the weak residue classes $a = 3, 4$, the constant term at $x = 0$ vanishes, giving $\Psi_a(z) = e^{-z} G_a(e^{-z})$. Each amplitude is analytic for $|e^{-z}| < 1$ and continues analytically to the small box used here. The error bounds below do not rely on this factorization alone: they use the exact real-axis Φ_a and the x factor in the derivatives.

Here $b_1 = b_4 = 1/150$ and $b_2 = b_3 = -11/150$. For conjugate terms use $u/(1-u) + u^{-1}/(1-u^{-1}) = -1$ to obtain

$$D_\ell(z) = -z/30 + ze^{-z}C_\ell(e^{-z}), \quad C_\ell(y) = \frac{1}{2} \sum_{j=1}^4 b_j \frac{\zeta^{\ell j}}{1 - \zeta^{\ell j} y}. \quad (9.27)$$

Consequently $\Theta_a = -z\Psi_a/30 + ze^{-z} \sum \zeta^{-a\ell} A_\ell C_\ell$. The first Euler–Maclaurin correction therefore retains the small factor in the weak residue classes. With $Z = (\tau^2 + h^2)^{1/2}$ and $p_a = |\Phi_a(e^{-\tau})|$,

$$D_* = Z \left(\frac{1}{30} + \frac{2x}{25d} \right), \quad |\Theta_a| \leq \frac{Z}{30} (p_a + 4ha_1) + \frac{8MZx}{25d} =: T_a. \quad (9.28)$$

We used $\sum |b_j| = 4/25$. For each root the remaining exponentiation error satisfies

$$\left| e^{D_\ell/n + \varepsilon_{\ell,n}} - 1 - D_\ell/n \right| \leq \frac{(3400 + D_*^2/2)e^{D_*/N + 3400/N^2}}{n^2}. \quad (9.29)$$

No x factor is claimed for this second-order remainder; it is handled by the explicit n^{-2} bound.

9.3. Gaussian replacement and symmetric integration

Let v_x be the variance of the real distribution $\Pr(J = j) \propto e^{-j\tau x}$, and put $V = \int_0^1 x^2 v_x dx$ and $W_r = \int_0^1 x^r v_x dx$ for $r = 3, 4$. Then $V \leq 4/3$. Direct interval integration in §9.4 gives $V > 1/80$ throughout.

For $Y = J - \mathbb{E}J$, we have $|Y| \leq 4$ and $v_x \leq 4$. Translation preserves the absolute value of the characteristic function. Since $J - 2 \in [-2, 2]$, for $|\omega| \leq 0.4$, $|\mathbb{E}e^{i\omega Y}| = |\mathbb{E}e^{i\omega(J-2)}| \geq \cos(2|\omega|) \geq \cos 0.8 > 0.69$. Also $|\mathbb{E}Y e^{i\omega Y}| \leq 0.4v_x$, and the second, third and fourth absolute-moment bounds are $v_x, 4v_x, 16v_x$. Substitution into the third and fourth logarithmic derivatives gives the coefficients

$$4/.69 + 4.8/.69^2 + 2.048/.69^3 < 23, \quad 16/.69 + 37.6/.69^2 + 30.72/.69^3 + 9.8304/.69^4 < 240.$$

Hence $|R'''| \leq 23W_3$ and $|R^{(4)}| \leq 240W_4$. The inequality $1 - \cos y \geq (1/2 - 1.6^2/24)y^2$ for $|y| \leq 1.6$ gives $\Re F_n \leq -\beta nVt^2$, where $\beta = 0.39$. Under the saddle condition $-R'(\tau) = m/(5n^2)$, define

$$P_{\text{bulk}} = \frac{e^{nR(\tau) + m\tau/(5n)}}{5\sqrt{2\pi V} n^{3/2}}, \quad F_n(t) = n(R(\tau - it) - R(\tau) + iR'(\tau)t), \quad G_0 = -nVt^2/2.$$

Writing $D = F_n - G_0$ and $C_3 = inR'''(\tau)t^3/6$ gives $|D| \leq (23/6)nW_3|t|^3$ and $|D - C_3| \leq 10nW_4t^4$. The second-order remainder formula, integrated along the segment between exponents, yields

$$|e^{F_n} - e^{G_0} - C_3 e^{G_0}| \leq \left(10nW_4t^4 + \frac{529}{72}n^2W_3^2t^6 \right) e^{-\beta nVt^2}. \quad (9.30)$$

Write $A(t) = \Psi_a(\tau - it)$, $A_0 = \Phi_a$, and $A_1 = A'(0)$. Then

$$\begin{aligned} A(t)e^{F_n} - A_0e^{G_0} &= A_0(e^{F_n} - e^{G_0} - C_3e^{G_0}) \\ &\quad + (A(t) - A_0)(e^{F_n} - e^{G_0}) + (A(t) - A_0 - A_1t)e^{G_0} + (A_0C_3 + A_1t)e^{G_0}. \end{aligned} \quad (9.13)$$

The last term vanishes exactly on a symmetric interval. For the others use the change bound $4a_1|t|$ and the quadratic remainder $4a_2t^2$. Integrating the even Gaussian moments over the real line and dividing by $\sqrt{2\pi/(nV)}$ gives the error coefficient

$$H_a = \frac{1}{\sqrt{2\beta}} \left[p_a \left(\frac{240W_4}{32\beta^2V^2} + \frac{2645W_3^2}{192\beta^3V^3} \right) + 4 \left(\frac{23a_1W_3}{8\beta^2V^2} + \frac{a_2}{2\beta V} \right) \right]. \quad (9.31)$$

The first bracket is multiplied by the combined main term p_a , rather than 4. The extracted first Euler–Maclaurin correction contributes at most $T_a/(n\sqrt{2\beta})$. Define the remaining coefficient by

$$U = \frac{4M}{\sqrt{2\beta}} (3400 + D_*^2/2) e^{D_*/N + 3400/N^2} \quad (9.32)$$

Its error is at most U/n^2 .

Cover the intermediate angles $[0.4, 1.2]$ by sixteen intervals $[u, v]$ of width $1/20$. On each interval set $c = \frac{1}{2}(\sin(2v)/(2v))^2$, so that $1 - \cos y \geq cy^2$ for $|y| \leq 4v$. Interval verification gives $cu^2 > 0.052$ on all sixteen intervals. The profile variance lower bounds directly give $0.052V > 0.00074$ and $Vh^2/2 > 0.001$. With the updated prefactor 34:

Part	Normalized upper bound
Minor arcs	$15n^{3/2}e^{-n/1000}$
Intermediate major-arc angles	$34\sqrt{n}e^{-37n/50000}$
Extension to the full Gaussian	$8e^{-n/1000}$

Each is less than $1/n$ for $n \geq N$. Compare logarithms at the threshold and use the derivative $c - \alpha/n > 0$ thereafter. Therefore

$$\left| \frac{c_n(m)}{P_{\text{bulk}}} - \Phi_{m \bmod 5}(e^{-\tau}) \right| \leq \frac{H_a + T_a/\sqrt{2\beta} + 3}{n} + \frac{U}{n^2}. \quad (9.33)$$

The Cauchy variable is $\theta = 2\pi\ell/5 + t/(5n)$, and $\sqrt{2\pi/(nV)}/(2\pi \cdot 5n)$ supplies the prefactor in P_{bulk} .

9.4. Interval verification over all parameters

Set $\tau_i = (11/2)i/128$ and $x_j = j/200$. On every complete rectangle enclose

$$v_x = \sum_{j < k} (k - j)^2 \frac{e^{-(j+k)\tau x}}{(\sum_{r=0}^4 e^{-r\tau x})^2}$$

Multiply by exact monomial integration weights and sum to enclose V, W_3, W_4 . Evaluate the phases from (7.2) on the same complete intervals. Retain a separate lower phase bound for each residue class and combine p_a with the error formula using interval arithmetic.

All 128 intervals verify positivity of the five phases, $V > 1/80$, and five error-to-phase ratios below 0.98: 1,408 comparisons in total. The variance lower endpoint is approximately 0.01423554 and the largest upper ratio approximately 0.37280682. This is an enclosure of all 25,600 rectangles, not extrapolation from samples. [certify_profiles.py](#) and [profile_certificate.json](#) record every interval. Since the $1/n$ and $1/n^2$ terms decrease for all $n \geq N$, the same comparison applies throughout that range.

10. Connection to all coefficients and the zero set

Let $n \geq 31147$. Section 6 covers $0 \leq m \leq \rho n^2$, where $\rho = -5R'(11/2)$. For $\rho n^2 \leq m \leq 5n^2$, the unshifted saddle-point equation has a solution with $0 \leq \tau \leq 11/2$, and the uniform comparison of §9.4 makes the error in (9.33) smaller than the signed main term. Thus every interior coefficient has strictly the desired sign. The right half follows by reciprocal symmetry.

The zero coefficients are precisely those in $\{5j + 3, 5j + 4 : 0 \leq j < n\} \cup \{7\}$, arising from agreement with the infinite product; $5n + 8, 5n + 9$, arising from $d_1 = 0$ in the first tail formula; and their reflections. Every other coefficient has a strict sign by the endpoint and interior estimates, so (1.2) follows as well.

11. Supporting interval certificates

11.1. Interval certificates

The accompanying code used interval arithmetic in mpmath 1.3.0 at 60 decimal digits to check the following:

- `certify_scalars.py`: 82 comparisons in total: 37 endpoint, five phase, and 40 interior comparisons.
- `certify_localization.py`: 421 comparisons, including the classification of every denominator and all 200 radial intervals.
- `certify_profiles.py`: all 128 intervals and 1,408 comparisons.
- `certify_resonant_gap.py`: integer verification of the polynomial identity for each difference and the basis for monotonicity, together with an enclosure of the uniform gap on 400 intervals.

Decimal inputs are supplied as strings or integer ratios, and the lower endpoint of each difference interval is required to be strictly positive. Conversion to ordinary floating-point numbers is not used for decisions. Version 0.6 removed an unused B -check from the previous version, and v0.7 checks the sixteen intermediate-arc intervals as G16_0–G16_15. Comparisons G17–G18 directly derive the intermediate-arc and Gaussian-tail decay rates from the lower variance bound; G20

handles the N -dependence of the prefactor; and E34 and G25 explicitly check the connection to the endpoint region.

The results are recorded in [scalar_certificate.json](#), [localization_certificate.json](#), [profile_certificate.json](#), and [resonant_gap_certificate.json](#). These do not constitute an independent formal verification of the analytic reasoning or of the library and code as a whole.

12. The fixed endpoint threshold

After a simultaneous search over several parameters, v0.7 regenerated the interval certificates at $N = 31147$. The upper endpoint-error bound is approximately (0.84662267), below the adopted value (0.85). The upper bound for the major-arc error-to-phase ratio is also less than one.

The limitation identified here is the endpoint condition at the fixed value $w_{\max} = 0.0013$:

$$31146 < \frac{C}{25w_{\max}^2} = 31146.6805557\dots < 31147. \quad (12.1)$$

[threshold_boundary_certificate.json](#) verifies both inequalities by interval arithmetic. Thus, with this endpoint condition unchanged, the integer threshold cannot be lowered to 31146 or below. This is not a proof of optimality over all parameter choices or reorganizations of the proof. Possibilities such as changing $w_{\max}$ further, improving the endpoint error formula, or using coverage of a different coefficient region remain open. The best value found in a search must be distinguished from an absolute mathematical lower bound.

13. Provenance and references

The analytic derivation grew from AI-assisted working notes and successive revisions of cancellation estimates and explicit constants. OpenAI Codex contributed mathematical derivations, implementations and exposition; parallel Sol subagents and suggestions from other AI systems contributed to proof, review and publication tasks. No personal author is named, and independent human peer review is not claimed. This edition preserves the mathematical core while selecting only material needed for the large- n result.

- [RR] NIST DLMF, §17.2(vi), [Rogers–Ramanujan identities](#).
- [Eta] NIST DLMF, §23.18, [Dedekind eta modular transformations](#).
- [Coth] NIST DLMF, §4.36.3, [partial fraction expansion of coth](#).
- [W21] Liuquan Wang, [Sign Changes of Coefficients of Powers of the Infinite Borwein Product](#), Proposition 3.6 and its proof.
- [WK22] Chen Wang and Christian Krattenthaler, [An asymptotic approach to Borwein-type sign pattern theorems](#), especially the discussion in §11 of cancellation for the third conjecture.

The Rogers–Ramanujan identities, the 5-dissection of the infinite product, the eta transformation, and the saddle-point method are established tools. A comprehensive literature search has not been completed concerning the novelty of the nonnegative-kernel identity, its application to the finite tail, the eventual theorem, the effective threshold, or the zero-set classification.